
Title:

MEDICARE OPIOID PRESCRIBING TRENDS (2017–2023)

**Research Question: "What prescribing patterns become
visible when opioid claims are normalized instead of using
raw counts?"**

DATA233

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DATA & METHODS

➤ CMS Medicare Part D (2017–2023)

➤ Variables:

➤ opioid claims

➤ provider type

➤ rural vs urban

➤ region

➤ long-acting opioid share

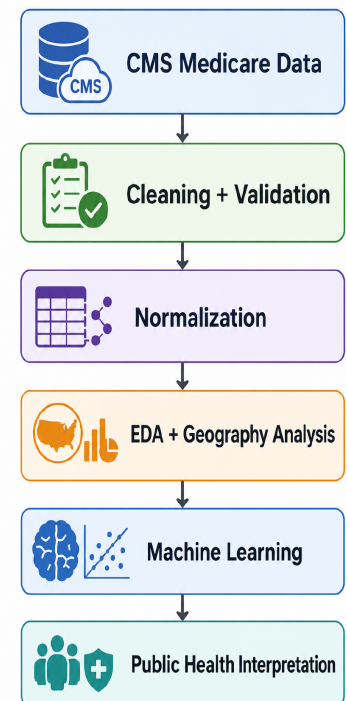
➤ Models:

➤ Logistic Regression

➤ Decision Tree

➤ Random Forest

Normalization: "Opioid claims per 1,000 total claims"

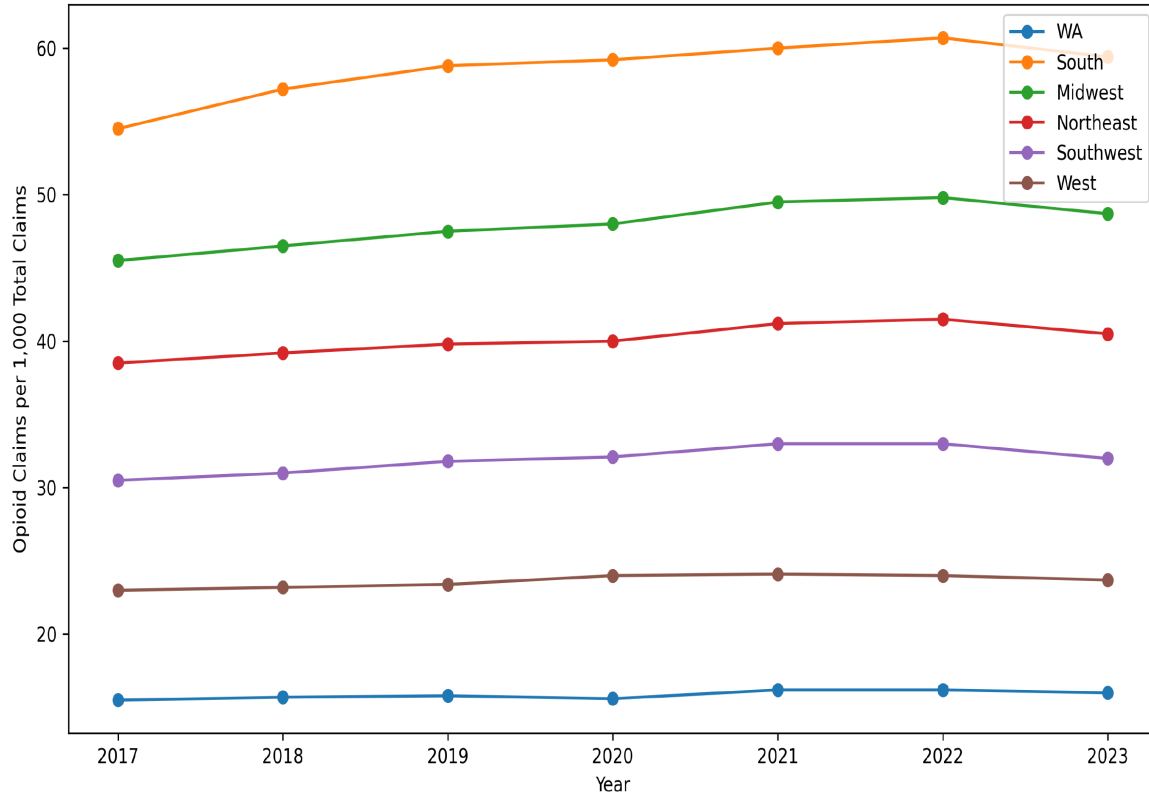


Key Point

Washington showed a steady decline in normalized opioid prescribing from 2017-2023, but remained higher than aggregated regional weighted rates in the final analysis.

REGIONAL FINDINGS

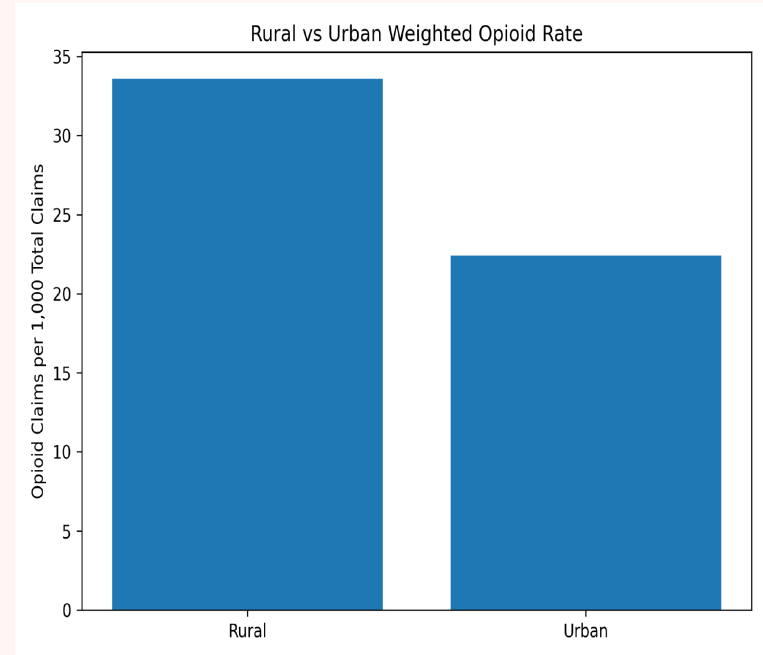
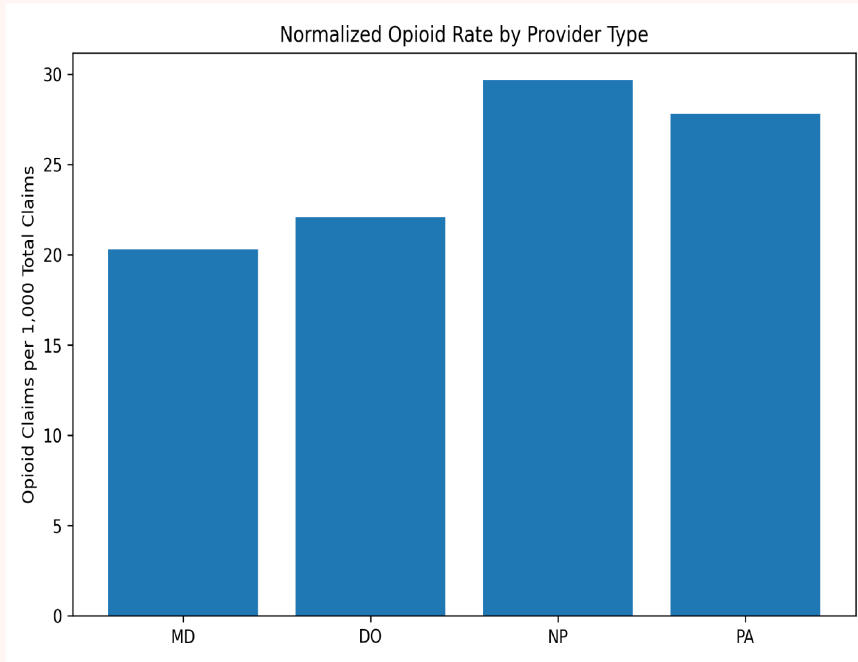
Washington vs U.S. Regions (Weighted Opioid Rate)



Key Point

Washington consistently showed the lowest normalized opioid prescribing rates.

PROVIDER TYPE + RURAL VS URBAN



➤ Key Findings

- **Nurse Practitioners and Physician Assistants showed higher normalized opioid rates.**
- **Rural providers prescribed more opioids per total claims than urban providers.**

Key Findings

- Long-acting opioids were a smaller overall share.
- Certain providers had disproportionately high long-acting shares.
- Random Forest performed best.

Important ML Framing

- Machine learning supports: screening, pattern identification, and provider risk analysis.
- Region and prescribing intensity contributed more strongly to predictions than credential alone.

➤ Key Findings

- Long-acting opioids were a smaller overall share.
- Certain providers had disproportionately high long-acting shares.
- Random Forest performed best.

➤ Important ML Framing

➤ Machine learning supports:

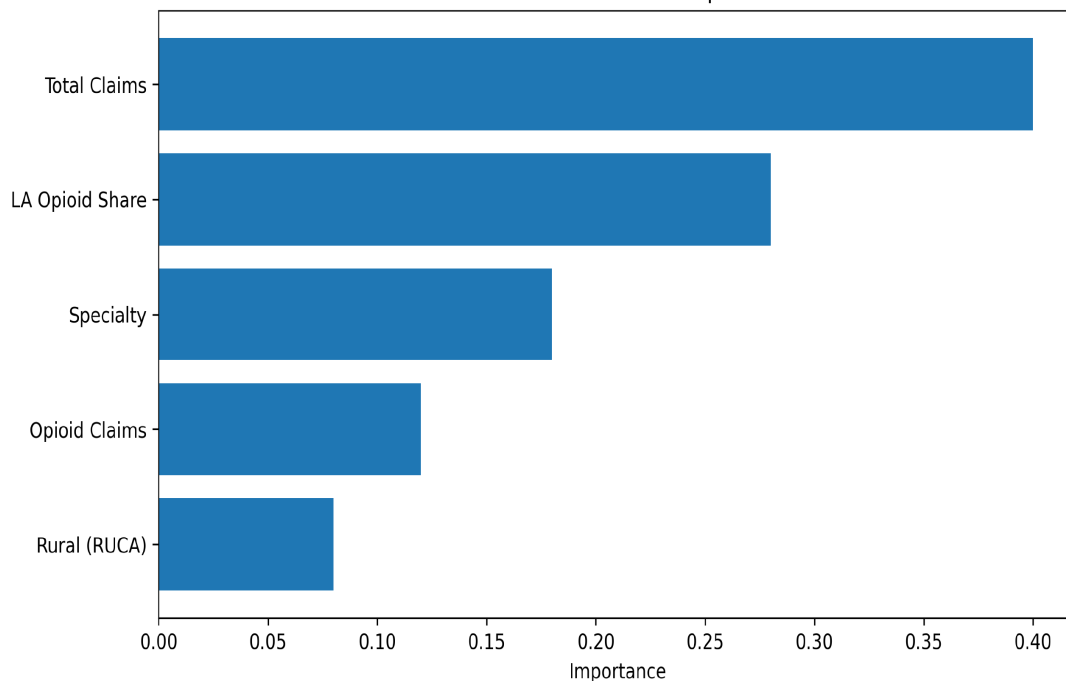
- screening
- pattern identification
- provider risk analysis

➤ Machine learning does NOT:

- diagnose misuse
- prove causation
- replace clinical judgment

LONG-ACTING OPIOIDS + ML RESULTS

Random Forest Feature Importance



LIMITATIONS + ETHICAL FRAMING

➤ Limitations:

- Claims data lacks patient clinical context.**
- Associations are not causation.**
- Medicare Part D only.**
- Provider-per-capita adjustments were unavailable.**

➤ Ethics:

- High prescribing does not automatically imply misconduct.**
 - Some providers treat higher-risk patient populations.**
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CONCLUSION + RECOMMENDATIONS

➤ **Main Conclusion:** "Normalized opioid rates reveal meaningful prescribing differences across provider types and regions that raw totals can hide."

➤ **Recommendations:**

- improve non-opioid pain management access
- expand rural treatment access
- continue monitoring long-acting opioid outliers
- use ML as a screening support tool

➤ **Final Takeaway**

➤ **"Normalization creates fairer comparisons and better identifies prescribing patterns for public health analysis."**
